5 Discrete-Time System Analysis using the z-Transform

5.1 The z-Transform

Put simply, the z-Transform is the discrete-time equivalent of the Laplace Transform, defined mathematically as

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n},$$

where $z = re^{j\theta}$. However, we'll never need to use this formula (or its inverse) specifically, and will refer to a table of transform pairs in most cases.

Similar to the Laplace Transform, the z-Transform is **linear**, meaning they obey the following property:

$$a_1x_1[n] + a_2x_2[n] \Leftrightarrow a_1X_1[z] + a_2X_2[z].$$

The unilateral z-Transform, defined as

$$X[z] = \sum_{n=0}^{\infty} x[n]z^{-n},$$

is going to be used most often here, since the signals we will be dealing with are largely causal, and the bilateral transform is not needed. If we *do* use the bilateral transform, the region of convergence of the z-Transform will become relevant.

5.1.1 Example:

Find the z-transform and the corresponding ROC for the signal $\gamma^n u[n]$.

First, recall that the given function represents a signal that grows or decays exponentially (depending on the value of γ), beginning at $n = 0$, where the signal's value $x[0] = 1$, and either decays or grows for $n > 0$.

Formulating the transform mathematically,

$$X[z] = \sum_{n=0}^{\infty} \gamma^n u[n]z^{-n}.$$

Since the unit step function will always be equal to $u[n] = 1$ for all values of $n \geq 0$, we may simplify the transform to

$$X[z] = \sum_{n=0}^{\infty} \left(\frac{\gamma}{z}\right)^n,$$

which reduces to a geometric series of the form

$$1 + x + x^2 + x^3 + ... = \frac{1}{1-x},$$

if $x < 1$.

Therefore, we may simplify the infinite sum produced by the z-transform formula to

$$X[z] = \frac{1}{1 - \frac{\gamma}{2}} = \frac{z}{z - \gamma}$$

as long as $|z| > |\gamma|$. This condition defines the *region of convergence* for the bilateral transform.

When using the unilateral z-transform, we will follow a very similar process to our use of the Laplace transform thus far, matching a given function to a transform pair or several transform pairs in the given table, potentially making changes to the original function to match it more closely to the table.

For the inverse z-transform, we'll once again use partial fraction expansion.

5.1.3 Example:

Find the inverse z-transform of

$$\frac{8z-19}{(z-2)(z-3)}.$$

Here, we'll expand the given function to

$$X[z] = \frac{3}{z-2} + \frac{5}{z-3}$$

and use the transform pair

$$\gamma^{n-1}u[n-1] \Leftrightarrow \frac{1}{z-\gamma}$$

to find the solution as

$$x[n] = [3(2)^{n-1} + 5(3)^{n-1}]u[n-1].$$

To get a function using $u[n]$ instead, we may use **modified** partial fraction expansion, using $X[z]/z$ instead:

$$\frac{X[z]}{z} = \frac{-19/6}{z} + \frac{3/2}{z-2} + \frac{5/3}{z-3}.$$

Using the transform pair

$$\gamma^n u[n] \Leftrightarrow \frac{z}{z-\gamma},$$

we may find the solution as

$$x[n] = -\frac{19}{6}\delta[n] + [\frac{3}{2}(2)^n + \frac{5}{3}(3)^n]u[n].$$

Another important correlation between the Laplace- and z-Transform is the **system transfer function**, defined as

$$H[z] = \sum_{n=-\infty}^{\infty} h[n]z^{-n}$$

and

$$h[n] \Leftrightarrow H[z].$$